实验报告：示波器及其应用.

一. 数据处理

(1) 校准测量. 校准信号(矩形波) $U_{p-p}=2V$ $f=1kHz$.

Y轴灵敏度(v/Div)	0.5 (V/Div)		1 (v/Div)		2 (v/Div)	
	Y轴格数(Div)	K(v/Div)	Y轴格数(Div)	K(v/Div)	Y轴格数(Div)	K(v/Div)
Y轴微调校准位	4.0	0.5	1.1	1.1	1.0	2.0

$K=\frac{U_{p-p}}{Y\text{轴格数}}$ (v/Div).

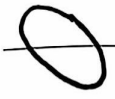
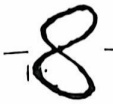
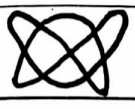
(2). 测量其他信号.

	Y轴灵敏度(v/Div)	Y轴格数(Div)	X轴灵敏度(mc/Div)	一个周期X轴格数(Div)
$f_1=200kHz$ $S=2V$	2.0	2.1	2×10^{-3}	2.4
$f_2=500Hz$ $S=6V$	2.0	6.0	1.0	2.1

	U_{p-p} (V)	U_e (V)	T(s)	f(kHz)
$f_1=200kHz$ $S=2V$	4.2	1.48	4.8×10^{-3}	208.3
$f_2=500Hz$ $S=6V$	12.0	4.24	2.1×10^{-3}	0.476

$U_e=\left(\frac{U_{p-p}}{2}\right)\times\frac{\sqrt{2}}{2}$ $f=\frac{1}{T}$.

(2). 李萨如图形测量正弦波频率.

图形	f_x (kHz)	f_y (kHz)	X轴交点 n_x	Y轴交点 n_y	$f_y:f_x$	$n_x:n_y$
	1	1	2	2	1:1	1:1
	2	1	2	4	1:2	1:2
	3	1	2	6	1:3	1:3
	2	3	6	4	3:2	3:2

Somewhere, something incredible, is waiting to be found.

扫描全能王 3亿人都在用的扫描App

(4) 振幅法测声速.

开始时 $f_1 = 35.735\ kHz$　结束时 $f_2 = 35.757\ kHz$　$t = 22^\circ C$

位置	I_1	I_2	I_3	I_4	I_5	I_6
读数(mm)	77.205	82.521	86.990	92.298 (x)	96.972	101.830
	I_7	I_8	I_9	I_{10}	I_{11}	I_{12}
	106.651	111.512	116.350	121.275	126.930 (x)	130.920

以位置角标为 X，读数为 y，设 $y = \hat{a} + \hat{b}x$，则 $\hat{b}$ 为 $\frac{\lambda}{2}$

易知 $\overline{x} = 5.5$　$\overline{y} = \frac{1}{10}\sum_{i=1}^{10} y_i = 99.3604$

$$\therefore \hat{b} = \frac{\overline{x}\cdot\overline{y} - \overline{xy}}{\overline{x}^2 - \overline{x^2}} = \frac{5.5 \times 99.3604 - 586.6}{5.5^2 - 38.5} = 4.8725\ (mm)$$

$$\lambda = 2\hat{b} = 9.7449\ (mm).$$

$$\hat{a} = \overline{y} - \hat{b}\overline{x} = 72.5617\ (mm)$$

$$r = \frac{\overline{xy} - \overline{x}\cdot\overline{y}}{\sqrt{(\overline{x^2} - \overline{x}^2)(\overline{y^2} - \overline{y}^2)}} = 0.99994$$

$$U_{b1}(\Delta\theta) = \frac{\Delta b_1}{\sqrt{3}} = \frac{0.005}{\sqrt{3}} = 2.887 \times 10^{-3}\ (mm).$$

$$U_{b2}(\Delta\theta) = \frac{\Delta b_2}{\sqrt{3}} = \frac{0.1}{\sqrt{3}} = 5.774 \times 10^{-2}\ (mm).$$

$$U_a(\hat{b}) = \hat{b}\cdot\sqrt{\frac{1}{n-2}\cdot\left(\frac{1}{r^2}-1\right)} = 0.01887\ (mm).$$

$$\therefore U(\hat{b}) = \sqrt{U_a^2 + U_{b1}^2 + U_{b2}^2} = 0.06008\ (mm).$$

$$\therefore U(\lambda) = 2U(\hat{b}) = 0.12016\ (mm).$$

最终结果表达为 $\lambda \pm U(\lambda) = (9.7 \pm 0.1)\ (mm)$.

CS 扫描全能王
3亿人都在用的扫描App

$f_0 = \frac{f_1+f_2}{2} = 35.746\,(kHz)$ $\Delta f = \frac{|f_1-f_2|}{2} = 1.1\times10^{-2}\,(kHz)$.

$\therefore U_b(f) = \frac{\Delta f}{\sqrt{3}} = 6.351\times10^{-3}\,(kHz)$

$U_a(f) = \sqrt{\frac{(f_0-f_1)^2+(f_0-f_2)^2}{2}} = 1.1\times10^{-2}\,(kHz)$

$U(f) = \sqrt{U_a(f)^2+U_b(f)^2} = 1.27\times10^{-2}\,(kHz)$

$v = \lambda f = 346.736\ m/s.$

$\frac{U(v)}{v} = \sqrt{(\frac{U(\lambda)}{\lambda})^2+(\frac{U(f)}{f})^2} = 0.0103$

$\therefore U(v) = 3.576\,(m/s)$

最终结果表达为 $v \pm U(v) = (347\pm4)\,(m/s)$

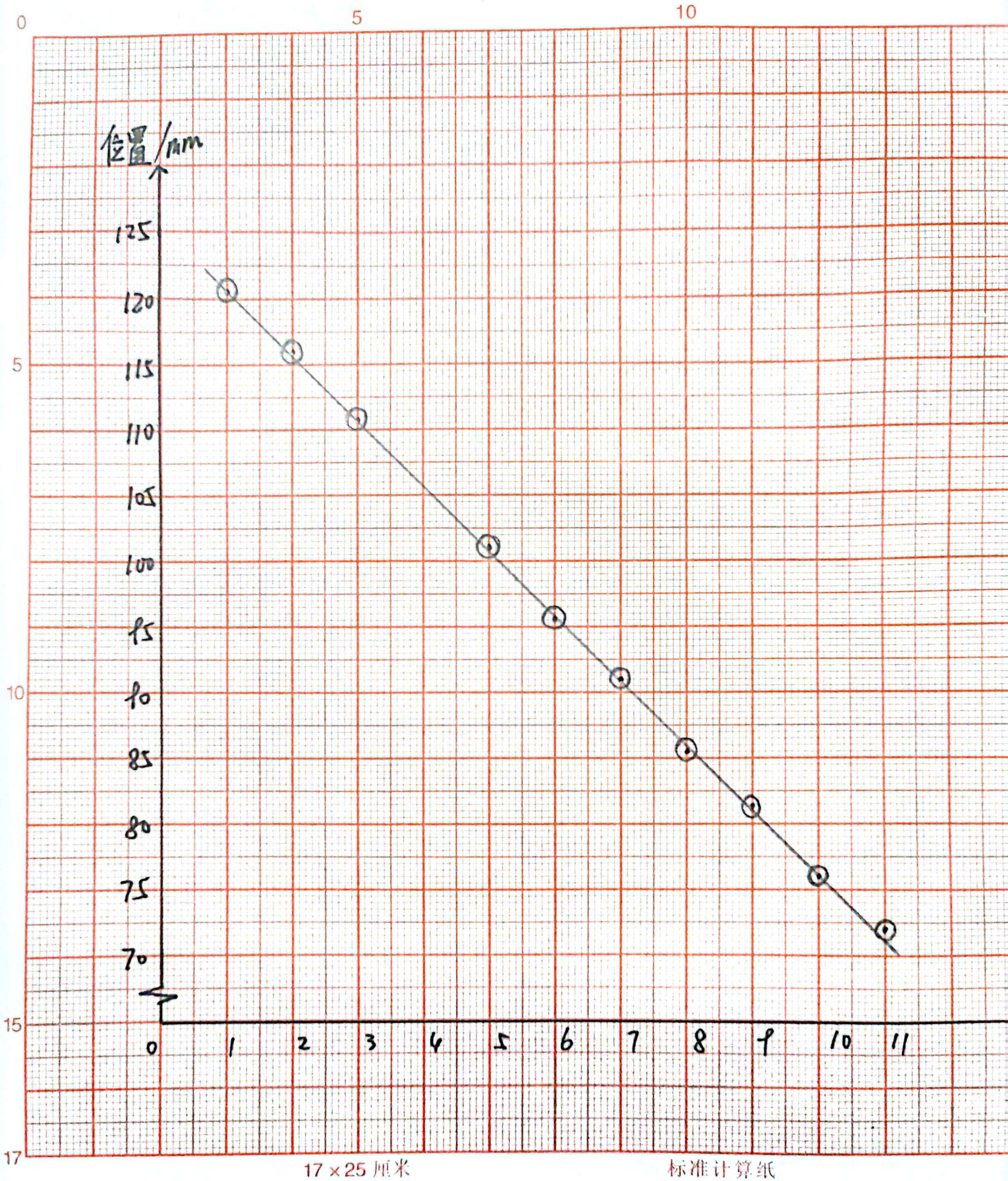
又 $\because v_{理论} = v_0\sqrt{1+\frac{t}{273}} = 343.7\ (m/s)$

$y = \frac{|v_{理}-v|}{v_{理}}\times100\% = 0.96\%.$

(5) 相位法测声速

开始时 $f_1 = 35.761\ kHz$ 结束时 $f_2 = 35.768\ kHz$ $t = 22^\circ C$

位置	I_1	I_2	I_3	I_4	I_5	I_6
读数(mm)	126.498	115.975	110.691	104.712	101.089	95.453
	I_7	I_8	I_9	I_{10}	I_{11}	
	91.28	85.491	81.42	76.229	71.812	



17×25 厘米 标准计算纸

扫描全能王
3亿人都在用的扫描App

在直线上选两点 (1.5, 118) 和 (10.5, 73.5).

解得 $b = -4.64$ (mm).

$\lambda = |2b| = 9.28$ (mm).

$\because f_0 = \frac{f_1 + f_2}{2} = 35.7645$ (kHz)

$\therefore v = \lambda f_0 = 353.4$ (m/s).

已知 $v_{理} = v_0\sqrt{1+\frac{t}{T}} = 343.7$ (m/s)

$\therefore \gamma = \frac{|v_{理} - v|}{v_{理}} \times 100\% = 2.8\%$.

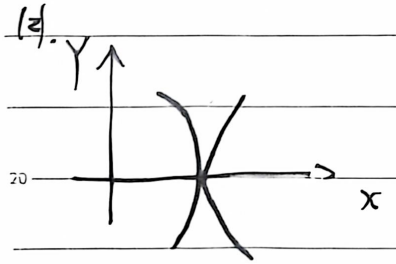
思考题

(1) 用 $T = 0.2$ ms 正弦电压做观察. 若出现 3 个完整波形,

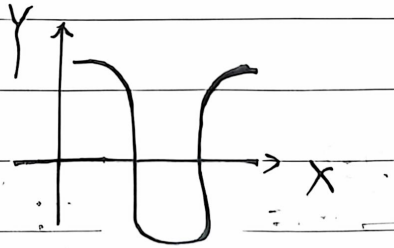
则 $T = 0.2 \times 3 = 0.6$ (ms)

事实上: T 在 0.6 ~ 0.8 ms 内均可观察到 3 个完整波形.

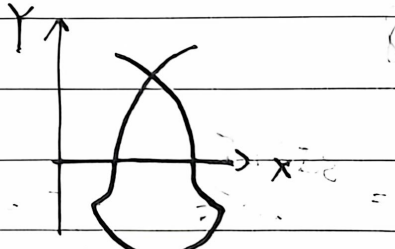
(2)



$f_y : f_x = n_x : n_y = 1 : 1$



$f_y : f_x = n_x : n_y = 2 : 1$



$f_y : f_x = n_x : n_y = 2 : 2$

$U_{p-p}=2V$ $U_p=1V$ $A=U_p$ $U_e=\frac{A}{\sqrt{2}}$

Y轴灵敏度 v/Div.	0.5 (v/Div)		1 (v/Div)		2 (v/Div)	
	格数 Y(Div)	K (v/Div)	格数 Y(Div)	K (v/Div)	格数(Div)	K(v/Div)
Y轴微调校准位.	4.0	0.5	1.1	1.1	1.0	2.0

	Y轴灵敏度(v/Div)	Y轴格数(Div)	X轴灵敏度(ms/Div)	一个T对应的X轴(Div)
$f_1=200kHz$ $A=2V$	2.0	2.1	2×10^{-3}	2.4
$f_2=500Hz$ $A=6V$	2.0	6.0	1.0	2.1

	U_{p-p}(V)	U_e(V)	T(s)	f(kHz)
$f_1=200kHz$ $A=2V$	4.2	1.48	4.8×10^{-3}	208.3
$f_2=500Hz$ $A=6V$	12.0	4.24	2.1×10^{-3}	0.476

图形	f_x(kHz)	f_y(kHz)	X轴交点 n_x	Y轴交点 n_y	$f_y:f_x$	$n_x:n_y$
	1	1	2	2	1:1	1:1
	2	1	2	4	1:2	1:2
	3	1	2	6	1:3	1:3
	2	3	6	4	3:2	3:2

李数是10.38

$f_1=$ 35.735 kHz $f_2=$ 35.737 kHz $t=$ 22 °C.

位置	I_1	I_2	I_3	I_4	I_5	I_6	I_7
读数(mm)	77.205	82.521	86.910	92.298	96.972	101.830	106.651

I_8	I_9	I_{10}	I_{11}	I_{12}
111.512	116.350	121.275	126.930	130.920.

Somewhere, something incredible. is waiting to be found.
扫描全能王 3亿人都在用的扫描App

$f_1 = 35.761\ kHz$　　$f_2 = 35.768\ kHz$　　$t = 22^{\circ}C$

位置	I_1	I_2	I_3	I_4	I_5	I_6
读数(mm)	120.488	115.875	110.681	104.712	101.081	95.453
	I_7	I_8	I_9	I_{10}	I_{11}	
	91.280	85.491	81.420	76.229	71.812	

扫描全能王
3亿人都在用的扫描App